React components
Module 2 React components

Package/Method	Description	Code Example
		1 2 3 4 5 6 7 8 9 10

React state	The state object is where you keep the component's property values.	class TestComponentextendsReact.component{
		constructor() {
		this.state= {
		id: 1,
		name: “John”
		age: 28
		};}
		render() {
		return ({this.state.name}{this.state.age}
		)}}
		Copied!

Props	Props is short for properties and it is used to pass data between React components.	1
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		7
		class TestComponentextendsReact.component{
		render() {
		return
		Hi {this.props.name}}
		//passing the props as examples to the test component
		TestComponentname= ‘John’
		TestComponentname= ‘Jill’
		Copied!

mounting	When a component instance is created and added to the DOM, these methods are invoked in the following order: constructor(),getDerivedStateFromProps(), render(),componentDidMount().	1
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		class Header extends React.Component {
		constructor(props) {
		super(props);
		console.log(”Inside the constructor”)
		}
		componentDidMount =()=> {
		console.log(”Inside component did mount”)
		}
		render() {
		console.log(”Inside render method”)
		return (
		The component is rendered
		)
		}
		}
		export default App
		Copied!

updating	When a component is updated,five methods are called in the same order: getDerivedStateFromProps(),shouldComponentUpdate(),render(), getSnapshotBeforeUpdate(),componentDidUpdate()	1 2 3 4 5 6 7 8 9 10 11 12 13

```
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17. 17
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21. 21
22. 22
```

```
1.  class App extends React.Component{
2. state = {counter: "0"};
3. incrementCounter = () => this.setState({counter:parseI
4. shouldComponentUpdate(){
5.  console.log("Inside shouldComponent update")
6.   return true;
7. }
8. getSnapshotBeforeUpdate(prevProps,prevState){
9. console.log("Inside getSnapshotBeforeUpdate")
10. console.log("Prev counter is" +prevState.counter);
11. console.log("New counter is" +this.state.counter);
12. return prevState
13. }
14. componentDidUpdate(){
15.   console.log("Inside componentDidUpdate")
16. }
17. render(){
18.   console.log("Inside render")
19.   return(<button onClick={this.incrementCounter}>Click
20. )
21. }}
22. export default App;
```

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```
1. 1
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3. 3
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22. 22
23. 23
```

Unmounting	When a component is removed or unmounted from the DOM, the componentWillUnmount() method enables us to run React code.

```
1.  import React from 'react';
2. class ComponentOne extends React.Component {
3. componentWillUnmount() {
4. console.log('The component is going to be unmounted');
5. }
6. render() {
7. return Inner component;
8. }
9. }
10. class App extends React.Component {
11. state = { innerComponent:<AppInner/>};
12. componentDidMount() {
13. setTimerout(()=>{
14. this.setState({ innerComponent:unmounted});
15. },5000)
16. }
17. render() {
18. console.log("Inside render")
19. return (
20. {this.state.innerComponent});
21. }
22. }
23. export default App;
```

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Changelog

Date	Version	Changed by	Change Description
20-10-2022	1.1	Sapthashree K S	Cheatsheet updated

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